

PERSONAL PROTECTIVE EQUIPMENT (PPE) SPECIFICATION

Sample engineering specification for Design//Human anthropometric coverage analysis

1. Product Identification

Field	Specification
Product	Industrial Safety Helmet
Model	DH-PPE-H01
Intended users	Adult industrial workers
Application	Construction and general industrial environments

2. Head and Fit Requirements

Parameter	Minimum	Maximum	Unit
Head circumference	540	620	mm
Head length	185	210	mm
Head breadth	145	170	mm
Vertical head height	215	250	mm
Helmet shell internal length	190	218	mm
Helmet shell internal breadth	150	178	mm
Adjustment range	540	620	mm

3. Fit and Safety Criteria

Criterion	Requirement
Coverage	The helmet must accommodate the specified head circumference range.
Clearance	Internal dimensions must provide sufficient clearance without excessive movement.
Adjustment	The retention system must allow adjustment across the specified range.
Stability	The helmet should remain stable during normal industrial movement.
Exclusion risk	Users outside the specified anthropometric range may require an alternative size or redesign.

4. Design Analysis Inputs

Analyze the stated dimensional limits against a representative adult human anthropometric dataset. Report estimated population coverage, potentially excluded measurement ranges, and design blind spots. Results are intended as a design-screening aid and should not replace formal PPE certification or applicable safety testing.

5. Revision Information

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